

-- Cox Regression - Sarcopenia Onset -----

i Definition : fnih
i Covariates : fixed
i Dairy : categorical
i Route : mice
i Interaction: none

-- MICE route - pooling across imputed datasets --

Detected 20 imputed datasets.
Fitting .imp = 1 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1117 rows | 32 events
Fitting .imp = 2 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1117 rows | 32 events
Fitting .imp = 3 / 20 ...
Events: 35 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1117 rows | 33 events
Fitting .imp = 6 / 20 ...
Events: 35 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1117 rows | 33 events
Fitting .imp = 7 / 20 ...
Events: 36 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1117 rows | 34 events
Fitting .imp = 8 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1117 rows | 32 events
Fitting .imp = 9 / 20 ...
Events: 36 sarcopenia cases out of 1117 participants

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1117 rows | 33 events
Fitting .imp = 11 / 20 ...
Events: 35 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1117 rows | 33 events
Fitting .imp = 12 / 20 ...
Events: 35 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1117 rows | 33 events
Fitting .imp = 13 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1117 rows | 32 events
Fitting .imp = 14 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1117 rows | 32 events

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1117 rows | 32 events
Fitting .imp = 17 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1117 rows | 32 events
Fitting .imp = 18 / 20 ...
Events: 35 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1117 rows | 33 events
Fitting .imp = 19 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Fixed dataset: 1117 rows | 32 events
Fitting .imp = 20 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

```
-- Events Per Variable (EPV) [mice (all imputations)]
```

```
i Events : 32
```

```
i Coefs : 10
```

```
i EPV : 3.2
```

```
-- C-index (Harrell's concordance) [mice (averaged)]
```

```
i C-index : 0.773 (95% CI: 0.702-0.843)
```

```
i SE : 0.0361
```

```
??? C-index >= 0.70 - good discrimination.
```

```
-- Global model tests [mice (averaged)]
```

```
i Likelihood ratio test :  $\chi^2(10) = 23.95$ ,  $p = 0.0081$  <- preferred
```

```
i Wald test :  $\chi^2(10) = 19.25$ ,  $p = 0.0386$ 
```

```
i Score (log-rank) test :  $\chi^2(10) = 23.53$ ,  $p = 0.00939$ 
```

```
i Total person-years (mean across imputations): NaN
```

```
-- Assumption checks [mice_imp1] --
```

```
-- 1. Proportional Hazards - cox.zph() | pass: global  $p > 0.05$ 
```

```
-- 2. Linearity - martingale (null model) vs continuous predictors | pass: smooth ??? linear
```

```
'geom_smooth()' using formula = 'y ~ x'
```

```
'geom_smooth()' using formula = 'y ~ x'
```

```
Linearity plots saved to
```

```
03_outputs/Cox/mice_fnih_fixed_categorical_27072026_1813
```

```
-- 3. Outliers - deviance & martingale residuals | pass:  $|dev| \leq 3$ 
```

```
-- 7. Log-log survival plot | pass: parallel lines across log(time)
```

```
Log-log plot saved to 03_outputs/Cox/mice_fnih_fixed_categorical_27072026_1813
```

```
-- 8. Cox-predicted vs KM - calibration by dairy quartile
```

```
Cox vs KM calibration plot saved to
```

```
03_outputs/Cox/mice_fnih_fixed_categorical_27072026_1813
```

```
-- Kaplan-Meier plot by dairy category [mice_imp1]
```

```
Ignoring unknown labels:
```

```
o colour : "Dairy quartile"
```

```
i Log-rank p-value: 0.904
```

```
KM plot saved to 03_outputs/Cox/mice_fnih_fixed_categorical_27072026_1813
```

```
-- Incidence rates by dairy quartile [mice]
```

```
quartile n_pts n_events person_years rate_per_1000py ci_low ci_high 1 Q1 268
```

```
7.3 19000.50 0.38 0.16 0.78 2 Q2 292 8.0 20690.00 0.39 0.17 0.76 3 Q3 273 9.3
```

```
19367.92 0.48 0.22 0.90 4 Q4 284 8.0 20241.30 0.40 0.17 0.78
```

```
Incidence table saved to
```

```
03_outputs/Cox/mice_fnih_fixed_categorical_27072026_1813
```

```
Model results saved to
```

```
03_outputs/Cox/mice_fnih_fixed_categorical_27072026_1813/mice_model_results.csv
```